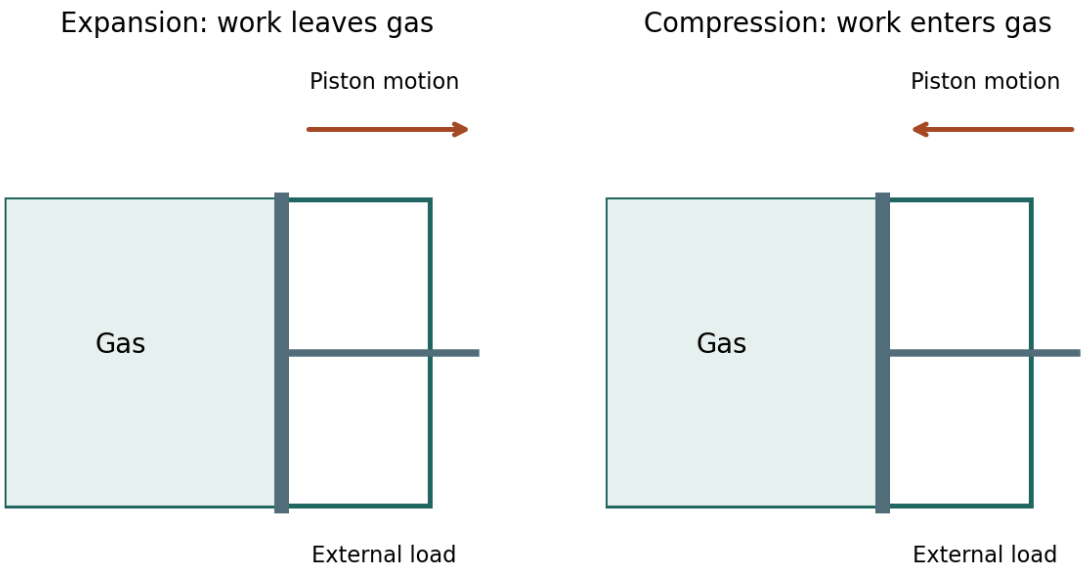
WS21 - Reference page

Keep this page beside the four question pages that follow.

Schematic expansion/compression against a load. Arrows indicate piston motion; identify the corresponding mechanical energy-transfer direction for the gas. Not a scale drawing.



panel	motion
expansion	outward
compression	inward

WS21 - Internal Energy, Heat, and Work

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS21 reference page preceding these question pages.

Objectives: Distinguish stored internal energy from heat and work transfers; explain expansion/compression; calculate constant-pressure boundary work and interpret energy arrows.

WS21-Q01 (2 points; foundation) Classify U, Q and W as a stored state property or a transfer during a process. Explain why the phrase “stored work” is misleading.

WS21-Q02 (2 points; foundation) A gas is compressed by an external piston. Is work BY the gas positive or negative in this unit? What sign describes work ON it?

WS21 - Internal Energy, Heat, and Work

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS21 reference page preceding these question pages.

WS21-Q03 (2 points; foundation) Which can occur in a rigid closed container? A. Pressure-volume boundary work from a moving wall. B. Energy input by an electrical interaction across the system boundary. Explain.

WS21-Q04 (2 points; foundation) Convert a 450 mL volume increase into liters. State the energy unit equivalent to kPa·L.

WS21 - Internal Energy, Heat, and Work

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS21 reference page preceding these question pages.

WS21-Q05 (3 points; application) A gas expands from 1.5 L to 4.0 L against constant external pressure 80 kPa. Calculate signed work by the gas.

WS21-Q06 (3 points; application) A gas is compressed from .0060 m^3 to .0020 m^3 against constant 125000 Pa. Calculate work BY the gas and the magnitude of work supplied ON it.

WS21 - Internal Energy, Heat, and Work

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS21 reference page preceding these question pages.

WS21-Q07 (3 points; application) Use the WS 21 piston diagram. Label the work arrow during expansion and compression as entering or leaving the gas. Explain why these arrows are not heat arrows.

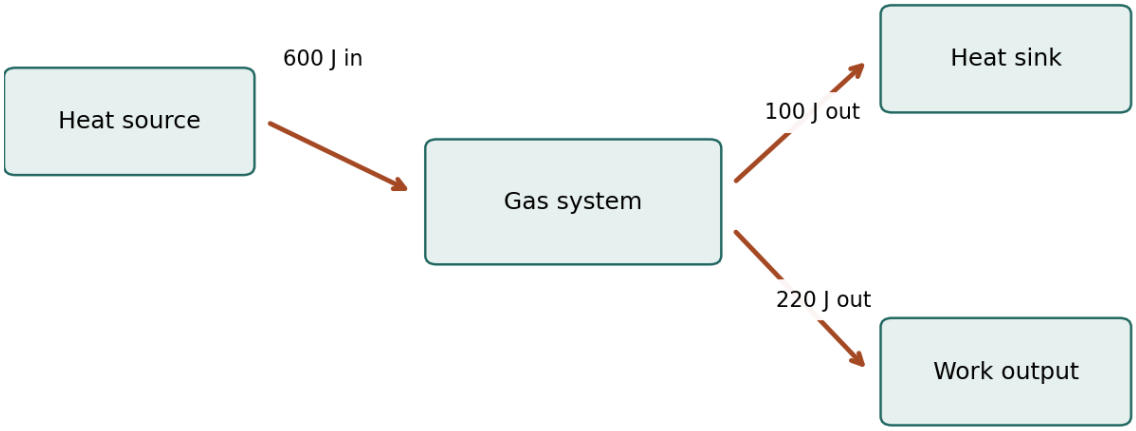
WS21-Q08 (3 points; challenge) Challenge: An ideal gas freely expands into vacuum inside an insulated rigid enclosure. Its occupied volume increases. Is W=P_extΔV positive? Determine W and Q, and explain why the volume change alone is insufficient.

WS22 - Reference page

Keep this page beside the four question pages that follow.

All arrows cross the boundary of one closed gas system. Whole-system motion and height changes are negligible. The heat arrows are separate transfers, not already net quantities.

WS22: separate transfers across one boundary



transfer	energy_J	direction
heat	600	in
heat	100	out
work	220	out

WS22 - First Law of Thermodynamics

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS22 reference page preceding these question pages.

Objectives: Apply ΔU=Q−W consistently; solve for any one unknown; translate work-on language; compare processes with the same endpoints and identify model limits.

WS22-Q01 (2 points; foundation) State the Unit 4 first law and define both transfer signs.

WS22-Q02 (2 points; foundation) A gas receives 90 J of heat and does 30 J of work. Find ΔU.

WS22 - First Law of Thermodynamics

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS22 reference page preceding these question pages.

WS22-Q03 (2 points; foundation) A source says 40 J of work is done ON the gas. What value belongs in W in ΔU=Q−W?

WS22-Q04 (2 points; foundation) True or false: If ΔU=0, both Q and W must be zero. Give a numerical counterexample or justification.

WS22 - First Law of Thermodynamics

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS22 reference page preceding these question pages.

WS22-Q05 (3 points; application) A closed gas releases 120 J as heat while surroundings do 200 J of work on it. Calculate ΔU and explain the direction of its change.

WS22-Q06 (3 points; application) A process has ΔU=−150 J and W=+250 J. Find Q and explain how heat input can accompany decreasing U.

WS22 - First Law of Thermodynamics

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS22 reference page preceding these question pages.

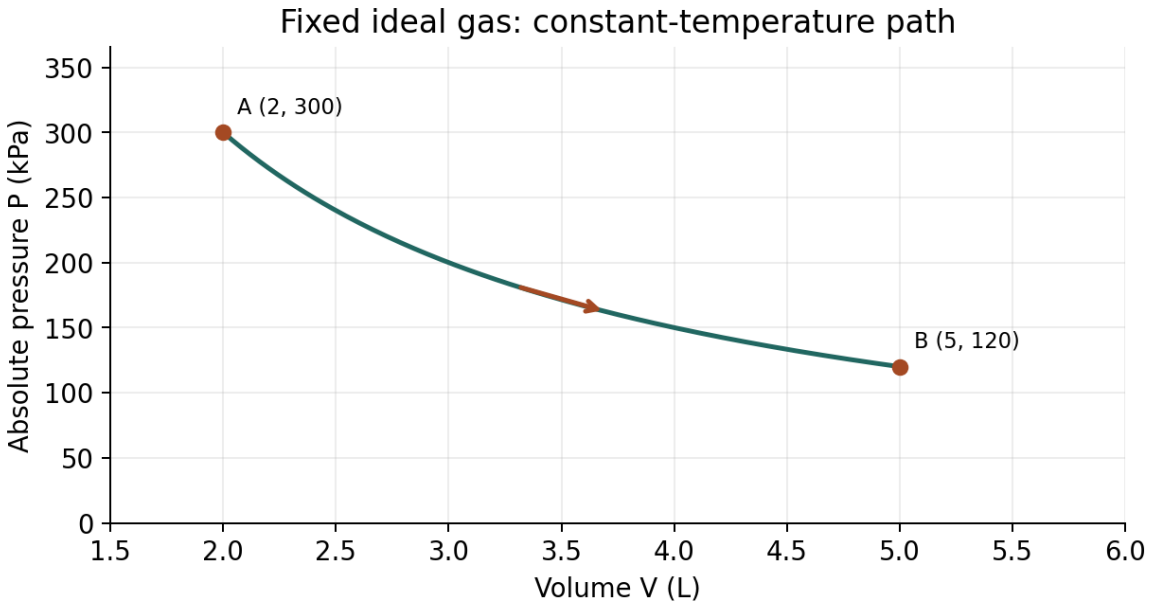
WS22-Q07 (3 points; application) Use the WS 22 energy ledger: 600 J heat enters, 100 J heat leaves and 220 J work leaves the gas. Calculate net Q and ΔU.

WS22-Q08 (3 points; challenge) Challenge: Routes A and B connect exactly the same initial and final states of a closed gas. A has Q=400 J and W=150 J. B has W=−50 J. Find ΔU for each and Q for B; explain why the heats can differ.

WS23 - Reference page

Keep this page beside the four question pages that follow.

Fixed amount of ideal gas on a slow, mechanically balanced isothermal expansion. The curve has PV=600 kPa·L. Do not calculate its curved-path work using a rectangle.



state	volume_L	absolute_pressure_kPa
A	2	300
B	5	120

WS23 - Isothermal Processes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS23 reference page preceding these question pages.

Objectives: Explain constant-temperature ideal-gas behavior; use Boyle’s law and Kelvin; interpret curved P–V paths; connect ΔU=0 to heat and work without calculating curved-path areas.

WS23-Q01 (2 points; foundation) Define isothermal. Does merely ending at the starting temperature prove that an entire process was isothermal?

WS23-Q02 (2 points; foundation) A fixed ideal gas doubles its volume isothermally. What happens to its absolute pressure?

WS23 - Isothermal Processes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS23 reference page preceding these question pages.

WS23-Q03 (2 points; foundation) For fixed ideal gas undergoing isothermal expansion, choose the correct statement: A. Q=0 always. B. ΔU=0 and Q=W. Explain.

WS23-Q04 (2 points; foundation) Why should an isothermal melting process not automatically be assigned ΔU=0 using the ideal-gas rule?

WS23 - Isothermal Processes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS23 reference page preceding these question pages.

WS23-Q05 (3 points; application) Fixed ideal gas starts at 300 kPa and 2.0 L. It expands isothermally to 5.0 L. Find final absolute pressure.

WS23-Q06 (3 points; application) A fixed ideal gas does 180 J of work during isothermal expansion. Find Q and ΔU.

WS23 - Isothermal Processes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS23 reference page preceding these question pages.

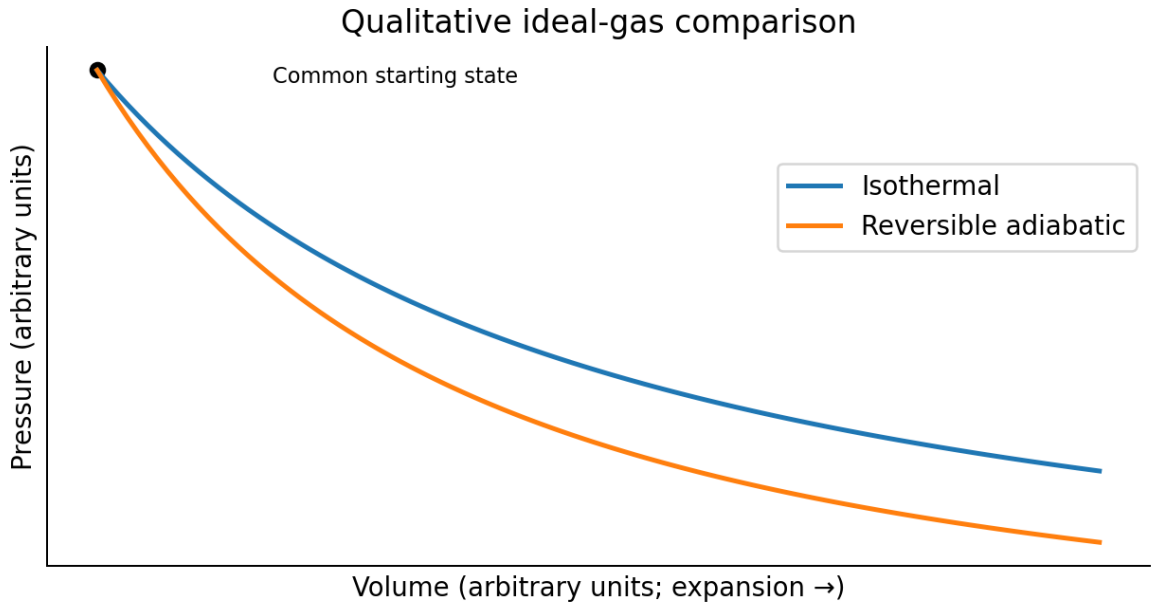
WS23-Q07 (3 points; application) Use the WS 23 curve. Verify PV at A and B, then state why P_A(V_B−V_A) is not a valid work calculation for this curved path.

WS23-Q08 (3 points; challenge) Challenge: Surroundings do 110 J of work ON a fixed ideal gas compressed isothermally. A student writes Q=+110 J because “compression heats gas.” Correct the signs and explain how temperature remains fixed.

WS24 - Reference page

Keep this page beside the four question pages that follow.

Conceptual comparison only: reversible ideal-gas expansion from the same state. Arbitrary-unit axes have no quantitative ticks. Use the relative curves, not invented numerical coordinates.



path	temperature_condition
isothermal	constant T
reversible adiabatic	Q=0 with positive work output

WS24 - Adiabatic Processes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS24 reference page preceding these question pages.

Objectives: Explain Q=0; apply the first law to expansion and compression; distinguish insulation, speed and reversibility; interpret a conceptual adiabatic P–V curve.

WS24-Q01 (2 points; foundation) Define adiabatic. Which quantity is constrained to zero?

WS24-Q02 (2 points; foundation) Adiabatic gas does 70 J of work. Calculate ΔU.

WS24 - Adiabatic Processes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS24 reference page preceding these question pages.

WS24-Q03 (2 points; foundation) Explain why rapid motion alone does not prove a process is exactly adiabatic.

WS24-Q04 (2 points; foundation) An insulated gas is compressed and 90 J of work is done on it. Give W_by and ΔU.

WS24 - Adiabatic Processes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS24 reference page preceding these question pages.

WS24-Q05 (3 points; application) Adiabatic ideal gas begins at 310 K with supplied C_U=10 J/K. It does 150 J of work. Find final temperature.

WS24-Q06 (3 points; application) Use the qualitative WS 24 graph. At the same enlarged volume, which path has lower pressure: reversible adiabatic or isothermal expansion from the same initial ideal-gas state? Explain.

WS24 - Adiabatic Processes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS24 reference page preceding these question pages.

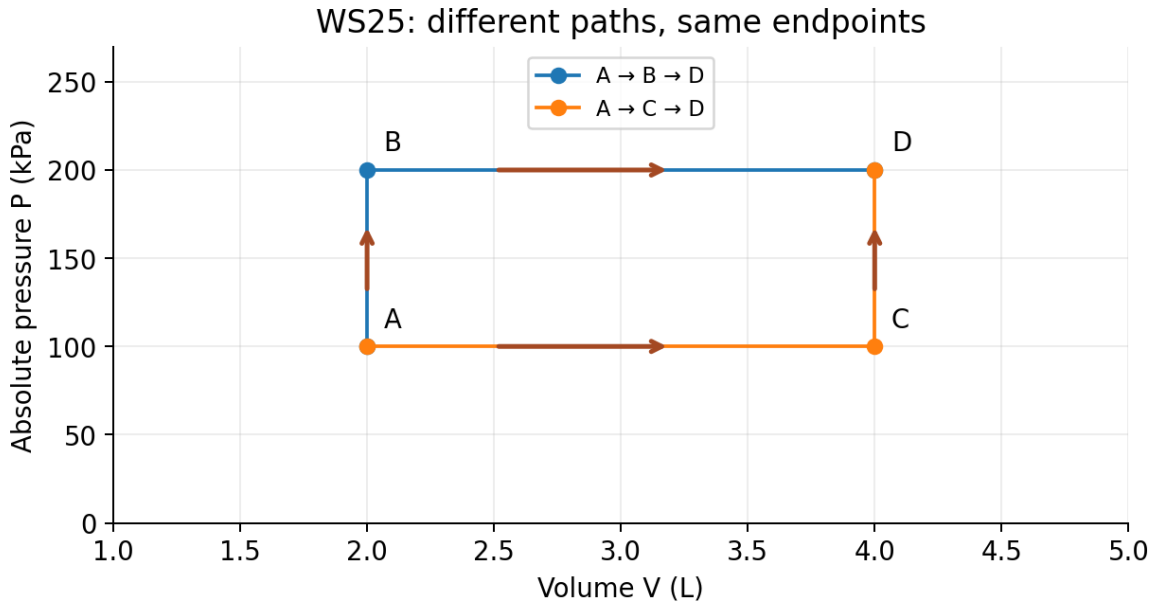
WS24-Q07 (3 points; application) An insulated ideal gas expands into vacuum with no resisting load. Compare its endpoint ΔU and ΔT with those of an adiabatic expansion that does positive work.

WS24-Q08 (3 points; challenge) Challenge: A gas with C_U=8 J/K falls from 300 K to 280 K while doing 200 J of work. Is the process adiabatic? Calculate Q and justify.

WS25 - Reference page

Keep this page beside the four question pages that follow.

Fixed gas, mechanically balanced paths, only PV work. A→B→D and A→C→D are two alternatives, not one closed cycle. Straight horizontal/vertical segments are exact model assumptions.



state	volume_L	absolute_pressure_kPa
A	2	100
B	2	200
C	4	100
D	4	200

WS25 - Isochoric and Isobaric Processes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS25 reference page preceding these question pages.

Objectives: Apply constant-volume and constant-pressure conditions; use Kelvin gas ratios; calculate work and first-law changes; compare four process types without confusing their constraints.

WS25-Q01 (2 points; foundation) Match isochoric and isobaric with (a) constant volume, (b) constant pressure. State which is a vertical line on a standard P–V graph.

WS25-Q02 (2 points; foundation) A fixed ideal gas at constant volume warms from 300 K to 330 K. If initial pressure is 100 kPa, find final pressure.

WS25 - Isochoric and Isobaric Processes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS25 reference page preceding these question pages.

WS25-Q03 (2 points; foundation) A rigid gas container receives 60 J of heat; no other work occurs. Find pressure-volume work and ΔU.

WS25-Q04 (2 points; foundation) Why can a constant-pressure warming ideal gas require more heat than its increase in internal energy?

WS25 - Isochoric and Isobaric Processes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS25 reference page preceding these question pages.

WS25-Q05 (3 points; application) Fixed ideal gas at constant pressure has V_i=3.0 L at 27.0 °C. It warms to 87.0 °C. Find V_f using 273.15 for the Celsius-to-Kelvin offset.

WS25-Q06 (3 points; application) Balanced isobaric expansion at 120 kPa takes gas from 2.0 to 4.5 L. If ΔU=180 J and only PV work occurs, find W and Q.

WS25 - Isochoric and Isobaric Processes

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS25 reference page preceding these question pages.

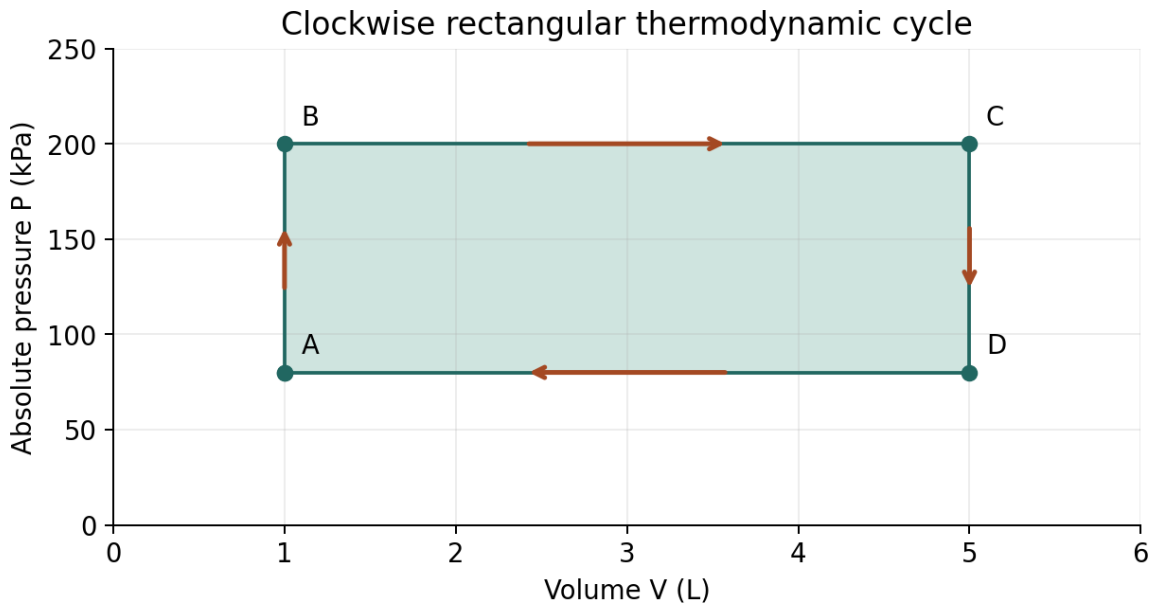
WS25-Q07 (3 points; application) Use the WS 25 paths. For A→B→D, calculate work on the vertical stage and horizontal stage. Compare total work with A→C→D.

WS25-Q08 (3 points; challenge) Challenge: A student says “the gas is insulated, so pressure is constant and no work occurs.” Identify each error and state what insulation actually permits you to conclude.

WS26 - Reference page

Keep this page beside the four question pages that follow.

Clockwise cycle A→B→C→D→A. Mechanically balanced, only PV work, closed gas returns to its original state. All segments are horizontal or vertical.



state	volume_L	absolute_pressure_kPa
A	1	80
B	1	200
C	5	200
D	5	80

WS26 - Thermodynamic Cycles

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS26 reference page preceding these question pages.

Objectives: Explain return to the initial state; calculate rectangular and triangular cycle work geometrically; use ΔU_cycle=0; interpret direction and net heat.

WS26-Q01 (2 points; foundation) What must return to its starting value in a complete thermodynamic cycle: U, total heat transferred, or total work transferred? Explain.

WS26-Q02 (2 points; foundation) On a P-vertical, V-horizontal diagram, what sign is net PV work by the gas for a simple clockwise loop?

WS26 - Thermodynamic Cycles

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS26 reference page preceding these question pages.

WS26-Q03 (2 points; foundation) A cycle receives net 300 J of heat. Determine ΔU_cycle and W_net.

WS26-Q04 (2 points; foundation) A gas moves vertically on a standard P–V plot. What is its PV work and why?

WS26 - Thermodynamic Cycles

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS26 reference page preceding these question pages.

WS26-Q05 (3 points; application) Use the WS 26 rectangle A→B→C→D→A. Find the work on each horizontal segment and the net work.

WS26-Q06 (3 points; application) A clockwise triangular cycle encloses a right triangle with base 3.0 L and pressure height 160 kPa. Find net work and net heat per cycle.

WS26 - Thermodynamic Cycles

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS26 reference page preceding these question pages.

WS26-Q07 (3 points; application) A cycle absorbs 900 J and rejects 650 J. Find W_net. If it completes 4 cycles each second at these values, find average mechanical output power.

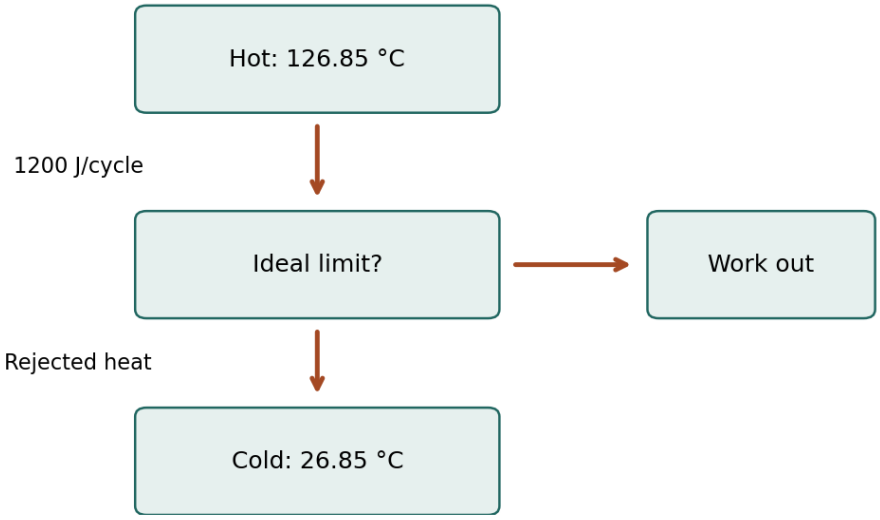
WS26-Q08 (3 points; challenge) Challenge: Reverse the WS 26 rectangular cycle. Find signed W_net and Q_net. Explain why negative work by the gas means work input, and why the diagram alone does not fully specify a refrigerator.

WS27 - Reference page

Keep this page beside the four question pages that follow.

Two-reservoir cyclic heat-engine model. Input is a positive heat magnitude of1200 J/cycle. Temperatures are given in Celsius and require conversion for the ideal bound.

WS27: reservoir temperatures and supplied input



quantity	value	unit
hot temperature	126.85	°C
cold temperature	26.85	°C
heat absorbed	1200	J/cycle

WS27 - Heat Engines and Thermal Efficiency

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS27 reference page preceding these question pages.

Objectives: Relate hot/cold reservoirs, work and heat rejection; compute actual efficiency and the Carnot upper bound; use Kelvin correctly and distinguish possible from proven performance.

WS27-Q01 (2 points; foundation) In an engine balance, Q_H and Q_C are defined as positive heat magnitudes. Write the cycle work equation and efficiency equation.

WS27-Q02 (2 points; foundation) An engine produces 240 J of work from 800 J heat input. Calculate efficiency as a fraction and a percent.

WS27 - Heat Engines and Thermal Efficiency

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS27 reference page preceding these question pages.

WS27-Q03 (2 points; foundation) Why does the efficiency denominator use heat absorbed rather than net heat for a complete cycle?

WS27-Q04 (2 points; foundation) Convert 76.85 °C and 26.85 °C to Kelvin for a reservoir-ratio calculation.

WS27 - Heat Engines and Thermal Efficiency

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS27 reference page preceding these question pages.

WS27-Q05 (3 points; application) An engine absorbs 1500 J per cycle and rejects 1050 J. Find work and thermal efficiency.

WS27-Q06 (3 points; application) An engine supplies 360 J/cycle at efficiency.40. Determine required Q_H and Q_C.

WS27 - Heat Engines and Thermal Efficiency

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Use the WS27 reference page preceding these question pages.

WS27-Q07 (3 points; application) Reservoirs are at 500 K and 300 K. Find the maximum ideal efficiency and decide whether a claimed 45% engine between them is possible.

WS27-Q08 (3 points; challenge) Challenge: Use the WS 27 reservoir diagram: hot 126.85 °C, cold 26.85 °C; Q_H=1200 J/cycle. Find maximum work output and minimum rejected heat. Explain why real work is lower.

WS28 - Second Law and Entropy

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

Objectives: Explain natural heat-flow direction, total entropy and irreversibility; distinguish reversible from merely repeatable; connect the first and second laws without advanced entropy mathematics.

WS28-Q01 (2 points; foundation) State the natural direction of net heat transfer between two objects with different temperatures and no external driving device.

WS28-Q02 (2 points; foundation) Does a refrigerator moving heat from cold food to a warmer room violate the second law? Explain.

WS28 - Second Law and Entropy

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

WS28-Q03 (2 points; foundation) Which is a more reliable entropy statement? A. Entropy is simply visible messiness. B. Entropy is a state property related to microscopic energy distribution, and total entropy of an isolated combination cannot decrease. Explain.

WS28-Q04 (2 points; foundation) State the unit of entropy and explain why it should not be confused with internal energy.

WS28 - Second Law and Entropy

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

WS28-Q05 (3 points; application) Supplied entropy changes for a complete isolated combination are ΔS_system=−3 J/K and ΔS_surroundings=+5 J/K. Find the total and assess consistency with the second law.

WS28-Q06 (3 points; application) A claimed isolated process has supplied changes +1 J/K for the system and−2 J/K for all its surroundings. Find total entropy change and assess the claim.

WS28 - Second Law and Entropy

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

ΔU=Q−W: Q into system positive; W by system positive. Show equations, signs and units.

WS28-Q07 (3 points; application) A working gas completes a cycle and returns to its initial entropy. Does this prove the cycle is reversible? Explain with the surroundings included.

WS28-Q08 (3 points; challenge) Challenge: A proposed cyclic engine takes 1000 J from 600 K, rejects 300 J to 300 K, and delivers 700 J work. Check both energy conservation and the two-reservoir efficiency limit. Which law rules it out?